

Kavya Bommineni

Data Scientist

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SUMMARY

Data Scientist with experience transforming complex data into scalable machine learning and AI solutions across healthcare and financial services. Specialized in predictive modeling, deep learning, NLP, Generative AI, and advanced machine learning with experience developing and optimizing models using XGBoost, LightGBM, PyTorch, TensorFlow, BERT, and Transformers. Experienced in building LLM-powered applications, RAG pipelines, prompt engineering, vector embeddings, and AI evaluation workflows for intelligent and context-aware solutions. Proven ability to improve model performance by 20%+ through feature engineering, experimentation, hyperparameter optimization, and model explainability. Strong collaborator skilled at translating complex business problems into innovative, production-ready AI/ML solutions.

SKILLS

- **Programming & Query Languages:** Python, SQL, Java
- **Machine Learning & Statistical Modeling:** Scikit-learn, TensorFlow, PyTorch, Keras, XGBoost, LightGBM, CatBoost, Feature Engineering & Feature Selection, Hyperparameter Tuning (GridSearchCV, Bayesian Optimization) Cross-Validation Techniques, Model Evaluation (ROC-AUC, F1, Precision-Recall, RMSE), Ensemble Methods, Dimensionality Reduction (PCA), Regression, Classification, Clustering, Anomaly Detection, A/B Testing, Time-Series Forecasting (ARIMA, LSTM, Prophet)
- **NLP, Generative AI & Agentic AI:** BERT, Hugging Face Transformers, Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), Agentic AI Systems (Autonomous Task Execution, Multi-step Reasoning), LangChain, Vector Databases (FAISS), Text Classification, Summarization, Named Entity Recognition (NER) Topic Modeling (LDA)
- **Data Engineering & Big Data:** Apache Spark, PySpark, Apache Kafka (Streaming Pipelines), Hadoop, MapReduce, Databricks, ETL & ELT Pipeline Development, Data Lakes & Warehousing, Snowflake, MongoDB, PostgreSQL, Data Modeling & Schema Design
- **Cloud, Fast Deployment & MLOps:** AWS (S3, EC2, Lambda, SageMaker), Azure Data Factory, GCP (Cloud Storage, Compute Engine), Docker, Kubernetes, REST API Development (Flask, FastAPI), MLflow (Experiment Tracking & Model Registry), CI/CD (Git, Jenkins, GitHub Actions), Model Deployment (Batch & Real-Time Inference), Model Monitoring & Drift Detection, Microservices Architecture, Serverless Deployment, Low-Latency Model Serving
- **Data Visualization & BI:** Power BI, Tableau, Matplotlib, Seaborn, Plotly, Excel (Advanced – Pivot Tables, Power Query), Streamlit, KPI Reporting & Dashboard Design
- **Workflow & Tools:** Jupyter Notebook, VS Code, Git & Version Control, Agile / Scrum, Data Storytelling

WORK EXPERIENCE

Own Your Voice Academy | Research And Development Intern June 2026 - Aug 2026

- Analyzed qualitative and quantitative datasets using Python, Streamlit, and Excel, performing data cleaning, preprocessing, SQL-based analysis, and EDA to identify trends and generate actionable insights.
- Developed interactive dashboards to analyze energy consumption, waste management, and sustainability metrics, translating complex datasets into actionable KPIs and visual insights.
- Collaborated with stakeholders to gather business requirements, define analytical objectives, and develop data-driven solutions using Python and SQL.
- Supported AI/ML initiatives through data preprocessing, exploratory analysis, feature engineering, and application of machine learning techniques to project datasets.

Suven Consultants & Technology Pvt. Ltd | Data Analytics Intern Apr 2022 - May 2022

- Processed and analyzed datasets using Python, Pandas, and NumPy, performing data cleaning, preprocessing, feature engineering, and exploratory data analysis.
- Analyzed meteorological patterns and trends to generate data-driven insights supporting machine learning applications and predictive analysis.
- Applied Python-based data processing and visualization techniques to support dataset analysis, integration, and interpretation.
- Evaluated data quality and analytical results to identify patterns, validate findings, and improve the reliability of insights for downstream machine learning workflows.

PROJECTS

AI Mine Assist – LLM + RAG Voice bot

- Developed an AI-powered voicebot using LLMs, NLP, and RAG to enable natural-language queries over complex mining regulations.
- Built semantic retrieval and RAG pipelines using LangChain, FAISS, and vector embeddings, integrating retrieved regulatory content with LLM-generated responses for context-aware information access.

Retail Shelf Product Detection using YOLOv5

- Developed an end-to-end computer vision pipeline for product detection across 5 retail categories and 200+ shelf images, applying preprocessing and augmentation techniques including Mosaic, MixUp, flipping, and scaling.
- Fine-tuned YOLOv5 using transfer learning, achieving 77.4% precision and 47.4% mAP@0.5, and evaluated performance using IoU, precision, and recall to identify detection gaps in dense and overlapping shelf scenarios.

Healthcare Cost Analytics Pipeline & BI Dashboard

- Built an SQL- and AWS-based ETL pipeline to extract, transform, and load healthcare data, performing data cleaning and modeling to support scalable analytics and BI reporting.
- Analyzed healthcare cost patterns using SQL and Excel and translated findings into interactive Power BI dashboards and visual reports to communicate data-driven insights and healthcare trends to stakeholders.

EDUCATION

University of New Haven	Jan 2025 - Aug 2026
Master of Science in Data Science	
• Coursework - Machine Learning, Artificial Intelligence, Deep Learning, Natural Language Processing(NLP), Data Engineering, Probability & Statistics, Linear Algebra	
Jansons Institute of Technology	July 2020 - May 2024
B.Tech Artificial Intelligence and Data Science	